CONFIDENTIAL · FOR TEACHERS ONLY

Mark Scheme

Grade 11 AS Level Physics · School Examination · 50 marks

SECTION A	SECTION B	SECTION C
15 marks	25 marks	10 marks

ASSESSMENT MAP

Topic / skill	Section A (MCQ)	Section B (Structured)	Section C (Practical)
Speed, velocity & acceleration	4, 5, 6, 7	16(d), 17	20(a)–(d)
Distance & displacement	2, 3	16(a)–(c)	—
Resolving & combining vectors	1, 8, 9, 10, 11	18	—
Coulomb’s law (2 and 3 charges; line & triangle)	12, 13, 14, 15	19	—
Experimental skills (data, graphs, uncertainty, improvements)	—	—	20(a)–(f)
Marks	**15 (30%)**	**25 (50%)**	**10 (20%)**

MARKING CODES

B1 Independent mark for a correct statement or value.

C1 Compensatory mark for correct working — awarded automatically if the final answer is correct.

A1 Accuracy mark for a correct final answer (and unit where asked).

ECF Error carried forward — allow a correct method using an earlier wrong value.

Accept answers to 2 or 3 significant figures unless the guidance says otherwise. Penalise a missing or wrong unit only where the answer line does not already give it.

Section A — answer key

1 mark each · 15 marks

Q1	Q2	Q3	Q4	Q5
D	**A**	**B**	**A**	**B**
Q6	Q7	Q8	Q9	Q10
C	**C**	**A**	**C**	**B**
Q11	Q12	Q13	Q14	Q15
D	**B**	**D**	**A**	**C**

Q	Key	Topic	Reasoning
1	**D**	Vectors/scalars	Displacement has magnitude and direction.
2	**A**	Speed & velocity	Speed = 400/80 = 5.0 m s^{-1}; back at start so displacement and velocity = 0.
3	**B**	Distance & displacement	Distance = 400 + 300 = 700 m; displacement = 400 − 300 = 100 m east.
4	**A**	Velocity from s–t graph	Gradient of R = (0 − 12)/(10 − 7) = −4.0 m s^{-1}.
5	**B**	Acceleration	a = (25 − 5.0)/8.0 = 2.5 m s^{-2}.
6	**C**	Motion graphs	Constant acceleration ⇒ straight sloping v–t line.
7	**C**	Average speed	Total distance/total time = 60/1.5 = 40 km h^{-1} (not the mean of 60 and 30).
8	**A**	Resolving vectors	θ is measured from the vertical, so vertical = F cos θ, horizontal = F sin θ.
9	**C**	Combining vectors	$\sqrt{(5.0^2 + 8.0^2)} = \sqrt{89} = 9.4$ N.
10	**B**	Combining/resolving	$\sqrt{(15^2 - 9.0^2)} = \sqrt{144} = 12$ m s^{-1}.
11	**D**	Resultant range	Resultant lies between 12 − 5 = 7 N and 12 + 5 = 17 N.
12	**B**	Coulomb's law (ratio)	$F \propto Q_1Q_2/r^2$: ×2 ÷ 2^2 = ×½.
13	**D**	Coulomb's law (2 charges)	$F = 8.99\times10^9 \times 2.0\times10^{-6} \times 3.0\times10^{-6} / 0.30^2 = 0.60$ N; unlike charges attract.
14	**A**	Coulomb's law (3 charges, line)	$9q/x^2 = q/(4d - x)^2$ ⇒ x = 3d from X, so P is d from Y.
15	**C**	Coulomb's law (triangle)	Two perpendicular forces of size F: $\sqrt{(F^2 + F^2)} = \sqrt{2}$ F.

Key distribution: A ×4, B ×4, C ×4, D ×3.

Section B — structured questions

25 marks

Part	Answer / marking points	Marks	Guidance
Question 16 — distance, displacement, speed and velocity			
(a)	Distance is a scalar (magnitude only) / total path length; displacement is a vector (has direction) / straight-line distance from start to finish.	**B1**	Any one valid difference.
(b)	600 + 800 = 1400 m	**A1**	
(c)	displacement = $\sqrt{(600^2 + 800^2)} = 1000$ m	**A1**	Allow 1.0 km.
	angle = $\tan^{-1}(800/600) = 53°$ north of east	**B1**	Allow 53.1°; allow bearing 037°.
(d)(i)	average speed = 1400 / 320 = 4.4 m s^{-1}	**A1**	Total time 120 + 200 = 320 s. Allow 4.375. ECF from (b).
(d)(ii)	average velocity = 1000 / 320 = 3.1 m s^{-1}	**A1**	Allow 3.125. ECF from (c).
	Total	**6**	
Question 17 — acceleration and the velocity–time graph			
(a)	Rate of change of velocity / change in velocity ÷ time taken.	**B1**	Not ‘change in speed’ / ‘velocity per time’.
(b)(i)	a = (20 − 0)/10 = 2.0 m s^{-2}	**A1**	
(b)(ii)	a = (0 − 20)/5.0 = −4.0 m s^{-2}	**A1**	Allow ‘deceleration of 4.0 m s^{-2}’. Sign or the word deceleration needed.
(c)	Displacement = area under graph: ½×10×20 + 20×20 + ½×5×20	**C1**	Allow trapezium: ½(20 + 35) × 20.
	= 100 + 400 + 50 = 550 m	**A1**	
(d)	average velocity = 550 / 35 = 16 m s^{-1}	**A1**	Allow 15.7. ECF from (c).
	Total	**6**	
Question 18 — resolving and combining vectors			
(a)	A scalar has magnitude only; a vector has magnitude and direction.	**B1**	Both parts needed.
(b)(i)	50 cos 35° = 41 N	**A1**	Allow 40.96.
(b)(ii)	50 sin 35° = 29 N	**A1**	Allow 28.7.
(c)	$v = \sqrt{(15^2 + 6.0^2)}$	**C1**	Pythagoras with both components.
	$= \sqrt{261} = 16$ m s^{-1}	**A1**	Allow 16.2.

	$\theta = \tan^{-1}(6.0/15) = 22°$ above the horizontal	**B1**	Allow 21.8°.
	Total	**6**	

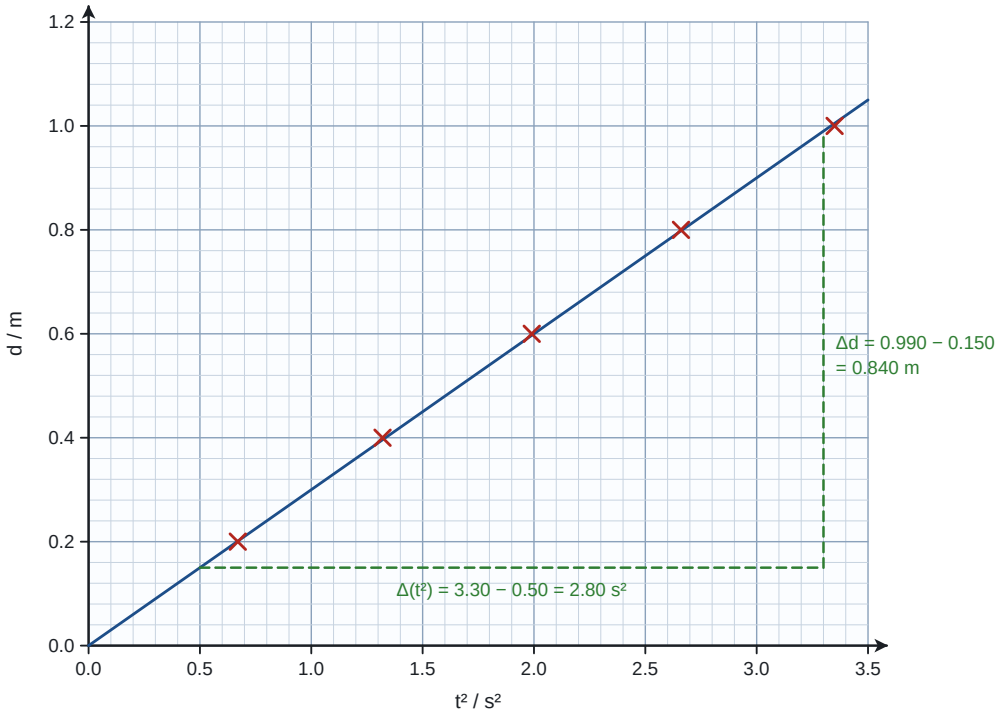
Question 19 — Coulomb's law: charges in a line and on a right-angled triangle

(a)	The force between two point charges is proportional to the product of the charges and inversely proportional to the square of their separation.	**B1**	Allow $F = kQ_1Q_2/r^2$ with symbols defined.
(b)(i)	$F_{AB} = 8.99\times10^{9} \times 3.0\times10^{-6} \times 2.0\times10^{-6} / 0.10^2 = 5.4$ N	**A1**	5.39 N; repulsive, so it acts on B away from A (to the right).
(b)(ii)	$F_{CB} = 8.99\times10^{9} \times 2.0\times10^{-6} \times 4.0\times10^{-6} / 0.20^2 = 1.8$ N	**A1**	1.80 N; attractive, so it acts on B towards C (to the right).
(b)(iii)	Both forces act in the same direction: 5.4 + 1.8 = 7.2 N, to the right (from A towards C).	**B1**	Magnitude and direction both needed. ECF from (i) and (ii). Answer 3.6 N (subtracting) scores 0.
(c)(i)	$F_X = 8.99\times10^{9} \times 3.0\times10^{-6} \times 2.0\times10^{-6} / 0.30^2 = 0.60$ N and $F_Y = 8.99\times10^{9} \times 3.0\times10^{-6} \times 2.0\times10^{-6} / 0.40^2 = 0.34$ N	**B1**	Both values needed for the mark (0.599 N and 0.337 N).
(c)(ii)	F_X drawn away from X (downwards); F_Y drawn towards Y (to the right). Resultant = $\sqrt{(0.60^2 + 0.34^2)} = 0.69$ N	**B1**	Arrows correct and resultant magnitude 0.68–0.69 N. ECF from (i).
	angle = $\tan^{-1}(0.60/0.34) = 61°$ below the line OY	**B1**	Allow 60°–61°. Allow 29°–30° measured from the downward vertical if clearly stated.
	Total	**7**	

Section C — practical skills

10 marks

Part	Answer / marking points	Marks	Guidance
Question 20 — practical skills: ball rolling down a ramp			
(a)	Mean t: 1.63 s (d = 0.80 m) and 1.83 s (d = 1.00 m)	**B1**	Both means correct, given to 2 d.p. (same as raw data).
	t^2: 2.66 s^2 and 3.35 s^2	**B1**	Both values correct to 3 s.f. ECF from mean t.
(b)	All five points plotted correctly to within half a small square.	**B1**	Points from the student's own table (ECF).
	Single thin straight line of best fit with points balanced either side; line passes through (or very close to) the origin.	**B1**	No 'dot-to-dot' or thick lines.
(c)	Triangle uses points on the line, hypotenuse at least half the length of the line, values read correctly, e.g. $\Delta d/\Delta(t^2) = 0.840/2.80$	**C1**	Do not allow use of table values unless they lie on the line.
	gradient = 0.30 m s^{-2}	**A1**	Accept 0.29 – 0.31.
(d)	gradient = a/2 ⇒ a = 2 × 0.30 = 0.60 m s^{-2}	**A1**	Accept 0.58 – 0.62. ECF from (c).
(e)(i)	$\Delta t = (1.66 - 1.61)/2 = 0.025$ s (≈ 0.03 s)	**A1**	Allow 0.025 or 0.03.
(e)(ii)	% uncertainty = 0.025/1.63 × 100 = 1.5 %	**A1**	Allow 1.5 – 1.8 % (1.8 % if 0.03 s used). ECF from (i).
(f)	Any one: use light gates connected to a timer/data-logger; video the motion and use frame-by-frame playback; use larger distances so times are longer (smaller % uncertainty); take more repeat readings and average.	**B1**	Must be a clear improvement to the timing. 'Be more careful' / 'use a better stopwatch' scores 0.
	Total	**10**	



Model answer for 20(b) and (c): points ×, best-fit line through the origin and a large gradient triangle (gradient = 0.840 / 2.80 = 0.30 m s^{-2}).